

Starts at 8:30 AM

**6th Workshop on
Memory-Centric Computing:
Logistics**

Ismail Emir Yuksel

ISCA 2026

28 June 2026

SAFARI

ETH zürich

Brief Self Introduction

- Ismail Emir Yuksel
 - Researcher @ SAFARI Research Group since September 2022
 - PhD @ ETH Zurich (Feb 2024 – ongoing)
 - ismail.yuksel@safari.ethz.ch (Best way to reach me)
 - <https://safari.ethz.ch>

- Research in:
 - Computer architecture, computer systems, hardware security
 - Memory systems
 - Hardware security, safety, predictability
 - Fault tolerance
 - Hardware/software cooperation
 - ...

Check the Website & Agenda



<https://events.safari.ethz.ch/isca26-MCCSys>

Organizers

■ **Ismail Emir Yuksel**

- PhD Candidate, SAFARI Research Group @ ETH Zürich

■ **F. Nisa Bostanci**

- PhD Candidate, SAFARI Research Group @ ETH Zürich

■ **Ataberk Olgun**

- PhD Candidate, SAFARI Research Group @ ETH Zürich

■ **Prof. Onur Mutlu**

- Professor of Computer Science at ETH Zurich
- <https://people.inf.ethz.ch/omutlu/index.html>

Agenda – First Session (08.30-10.00)

Time	Talk
08:30 – 08:40	Logistics/Welcome Ismail Emir Yuksel
08:40 – 10:00	Memory-Centric Computing: Recent Advances in Processing- using-DRAM Ismail Emir Yuksel
10:00 – 10:30	Coffee Break

Agenda – Second Session (10.30-12.00)

Time	Talk
10:30 – 11:00	Memory-Centric Scaling of LLM Decoding Through Contextual Sparsity Liu Liu
11:00 – 11:30	Simulation, Compilation, and Deployment Optimization for Near-DRAM Processing Architectures Zhenhua Zhu
11:30 – 12:00	DCC: Data-Centric Compilation of Machine Learning Kernels for Processing-In-Memory Architectures Peiming Yang
12:00 – 13:30	Lunch Break

Agenda – Third Session (13.30-15.30)

Time	Talk
13:30 – 13:50	Storage-Centric System Designs for Enabling Fast, Efficient, and Low-Cost Genomic and Metagenomic Analyses Nika Mansouri Ghiasi
13:50 – 14:15	In-DRAM Signature Generation Using Simultaneous Multiple-Row Activation: An Experimental Study of Off-The-Shelf DRAM Chips Umut Baser
14:15 – 14:40	Programmer-Transparent Near-Data Processing Using Multiple Compute-Capable Resources in Solid State Drives Rakesh Nadig
14:40 – 15:05	Efficient Memory Management and Access Mechanisms for CPU-PIM Co-Execution Systems Yilong Zhao
15:05 – 15:30	PuDGhost: Interference from Non-Operand Data in Processing-using-DRAM Daichi Tokuda
15:30 – 16:00	Coffee Break

Agenda – Fourth Session (16.00-18.10)

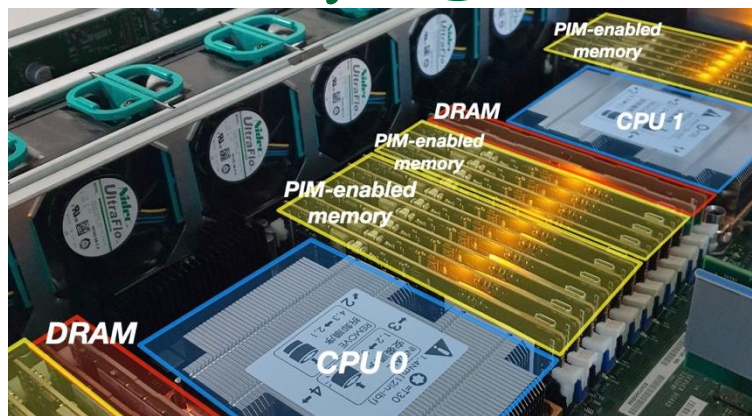
Time	Talk
16:00 – 16:25	Scaling the Memory Wall: Towards 3D-DRAM-based Accelerators for Efficient Generative Inference Prashant Nair
16:25 – 16:50	Revisiting Main Memory-Based Covert and Side Channel Attacks in the Context of Processing-in-Memory Nisa Bostanci
16:50 – 17:15	FPGA-Based Frameworks to Enable Fundamentally Better Computing Systems Ataberk Olgun
17:15 – 17:40	Acceleration of the Prefiltering and Alignment Phases in MMseqs2 Using the Processing-in-Memory Approach Mateusz Juszczak
17:40 – 18:05	Efficient Large Language Model Acceleration Using HBM-Based PIM Architectures Murali Annavaram
18:05 – 18:10	Closing Remarks Ismail Emir Yuksel

ACM International Conference on Supercomputing 2026

6-9 July 2026

Belfast, Northern Ireland, United Kingdom

7th Workshop on Memory-Centric Computing Systems



*More information &
call for papers:*



MCCSys

*A full-day workshop @ ICS 2026
6th July 2026 (Monday)*

Organizers: Ismail YukseI, Nisa Bostanci, Ataberk Olgun, Dr. Zhiheng Yue, Dr. Mohammad Sadrosadati, Dr. Geraldo F. Oliveira, Prof. Onur Mutlu

- Overview of PIM & PIM Taxonomy
- PIM in Memory & Storage
- Infrastructures for PIM Research
- Real-World PnM Systems
- PuM for Bulk Bitwise Operations
- Programming Techniques & Tools
- Research Challenges & Opportunities

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